# 从零开始移植Bootloader(极速版)

## 注意事项:

### 1.本文章只涉及如何移植并简单使用！！！

### 2.其中bootloder与ringbuffer知识请自行查询ai

### 3.其中所有的工程代码请自行查询我的github/gitee上进行拷贝学习

### 4.其中涉及路径最好不要包含中文

## 使用模块以及平台说明:STM32F407VET6(嘉立创的天空星)+HAL库

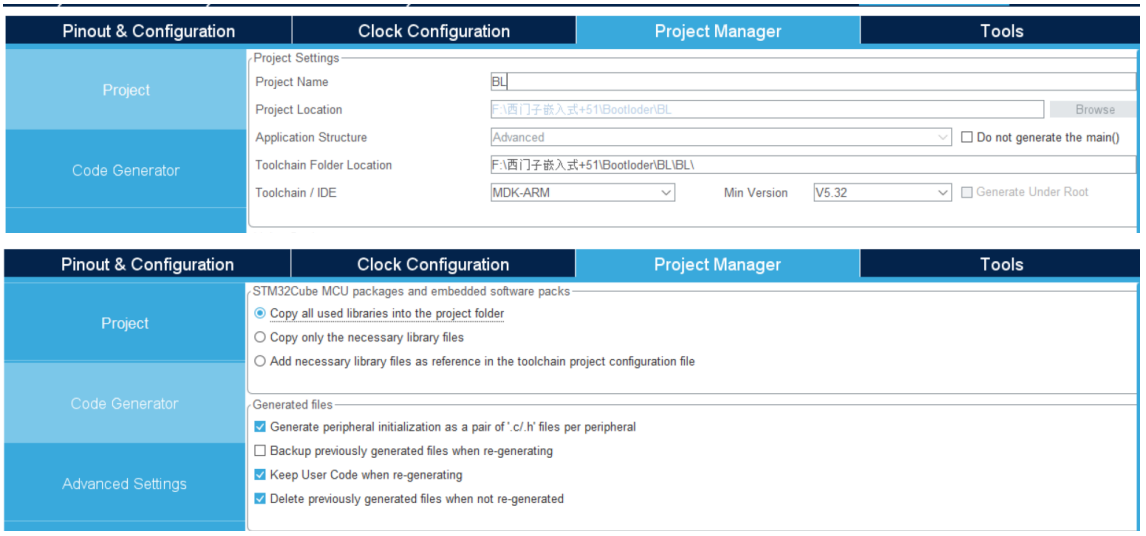
## BL工程创建与使用

### 1.文件夹创建

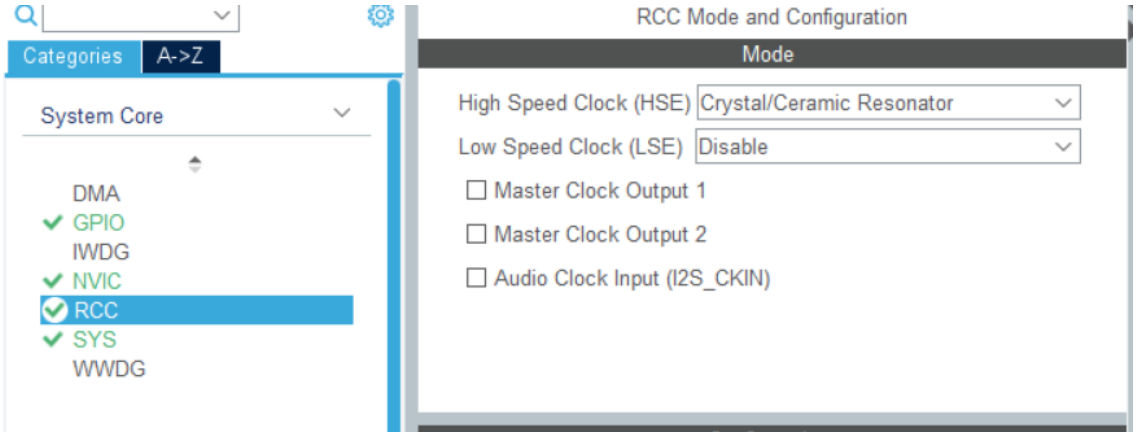
BL+BL_APP+APP+BL_min+Common(共5个文件夹)

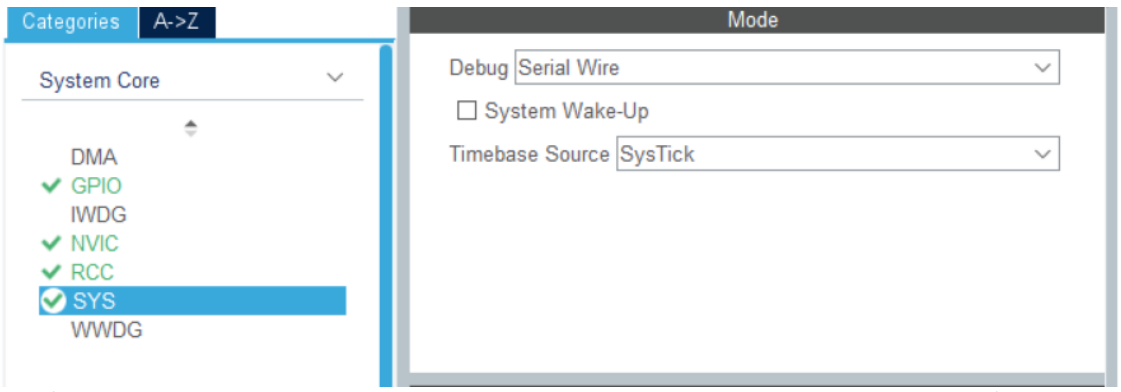
### 2.使用STM32CubeMX完成工程创建

1. 选择芯片STM32F407VET6
2. 修改文件名以及选择工程存放路径等操作(这是我的设置，仅供参考)

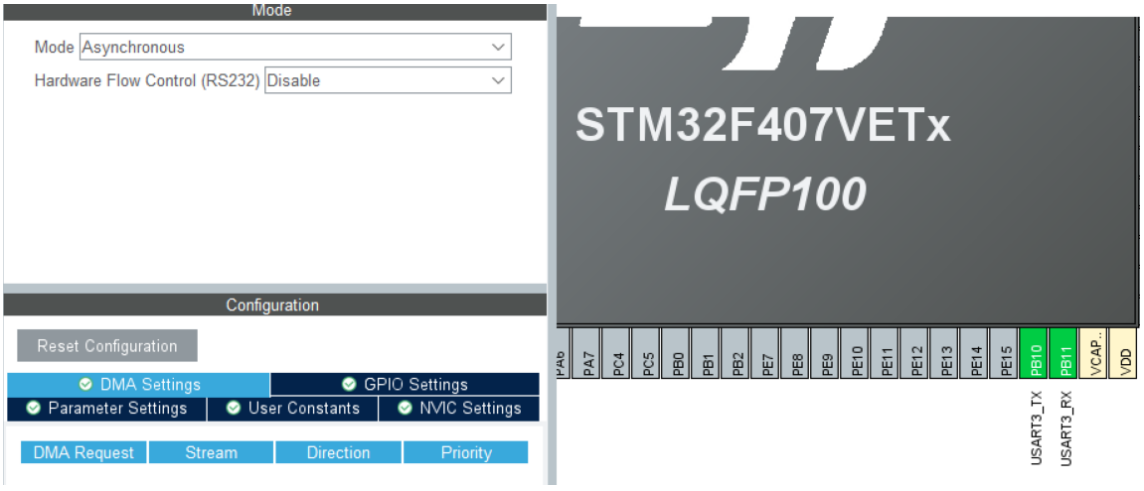



3. 将时钟与SYS一同打开

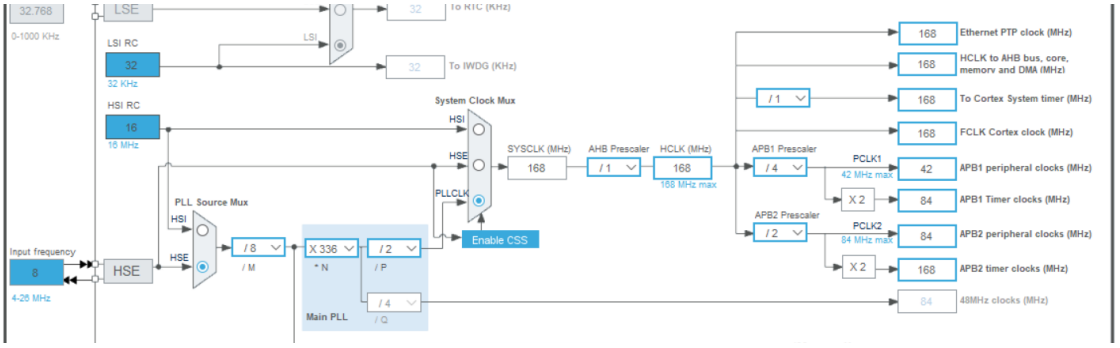



4. 打开**串口3**，并设置频率为**500000** Bit/s(串口引脚可自行选择)-该串口3就是我们的**OTA串口**



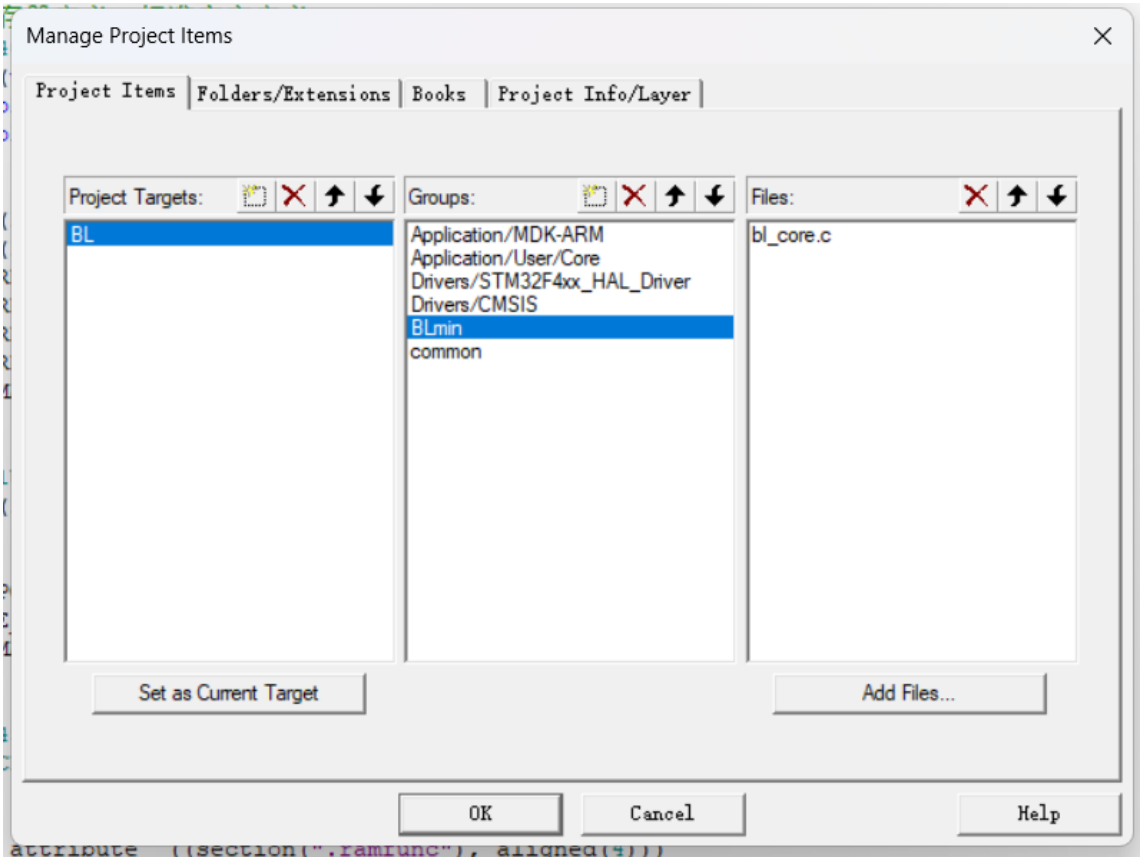

5. **设置时钟树(重点)**

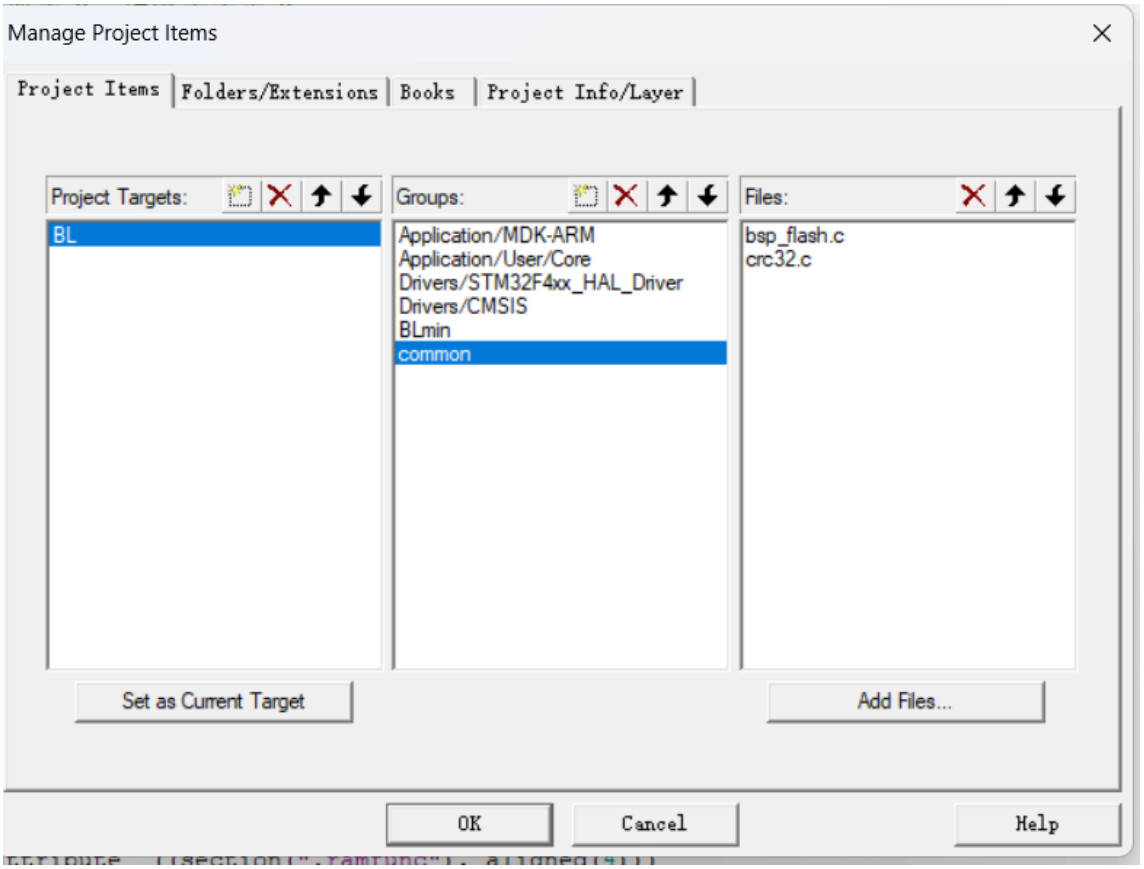



6. 生成工程代码

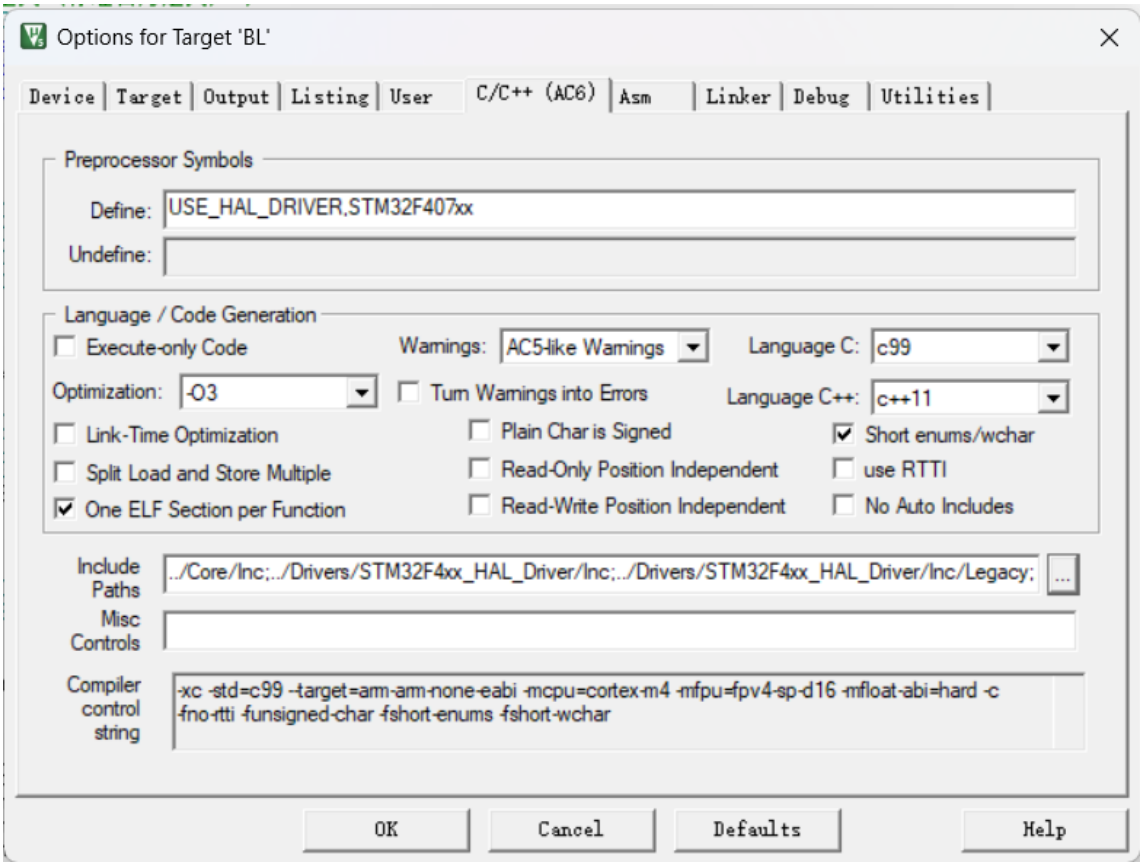
## 3.keil中文件的添加以及文件路径设置与代码添加

1. 添加文件如图所示




2. 点击魔法棒，添加头文件的路径(参考)-点击上面的Include Paths->然后添加刚刚创建好的BL_min和Common文件的路径即可(**注意:一定要点进文件夹**)

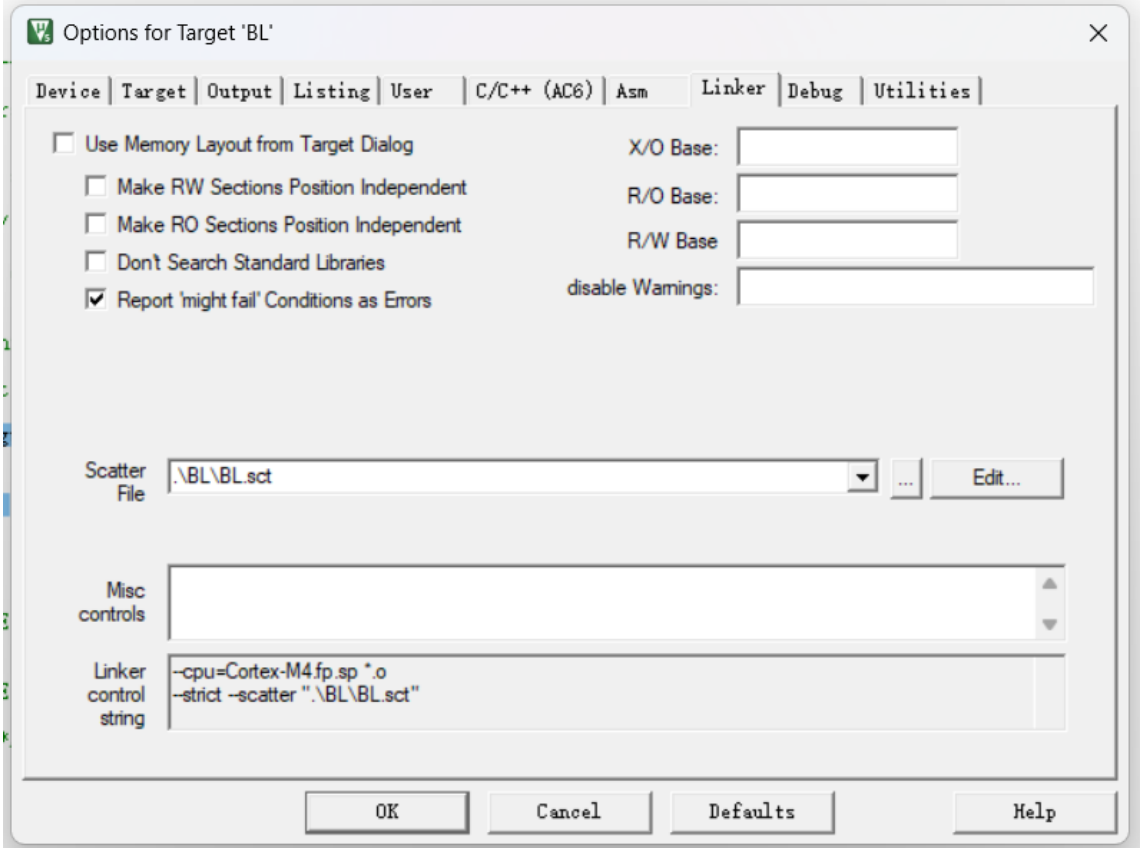



3. 打开main.c添加代码即可

```
/* USER CODE BEGIN Includes */
#include "bl_core.h"        //请添加在头文件引用中
/* USER CODE END Includes */
```

```
  /* Initialize all configured peripherals */
  MX_GPIO_Init();
  MX_USART3_UART_Init();
  /* USER CODE BEGIN 2 */
    bootloader_run();       //放置在硬件初始化后面
  /* USER CODE END 2 */
```

4. **首先将Linker中取消勾选(如图所示)->修改MDK_ARM文件夹下的BL.sct代码(修改或者直接复制我代码中BL.sct)->在Scatter File中选择修改好的BL.sct路径(重点)**




BL.sct代码

```
; *************************************************************
; *** Scatter-Loading Description File generated by uVision ***
; *************************************************************

LR_IROM1 0x08000000 0x0000C000 {    ; 48KB BL 空间（对齐 16KB，正确）
  ER_IROM1 0x08000000 0x0000C000 {
   *.o (RESET, +First)
   *(InRoot$$Sections)
   .ANY (+RO)
   .ANY (+XO)
  }
  RW_IRAM1 0x20000000 0x00030000 { .ANY (+RW +ZI) }  ; STM32F407 192KB RAM
  ER_IRAM_FUNC +0 {
   bsp_flash.o (+RO)                        ; Flash 代码放 RAM 执行（正确）
  }
}
```

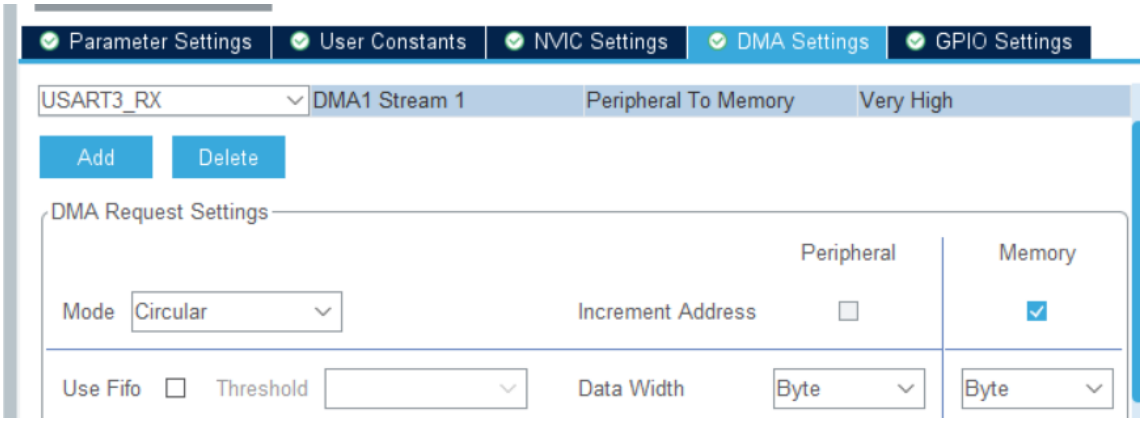
5. 编译下载->打开串口验证是否程序成功下载并运行(如图所示)

```
BL: --- log dump ---
BL: no valid APP, halt.
```

# 最小APP测试工程创建与使用

## 1.使用STM32CubeMX完成工程创建

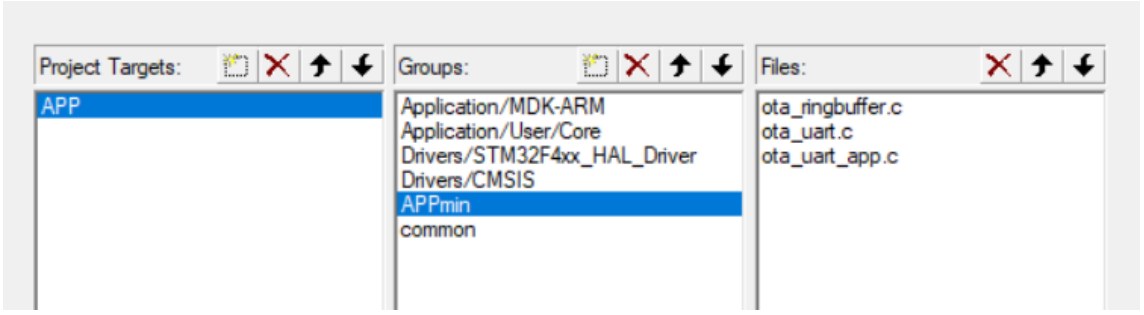
1. 创建项目流程与上方BL建立工程相同
2. 打开串口1(检验固件更新)-引脚自行选择
3. 选中串口3，对串口3的读取添加DMA(DMA设置如图所示)

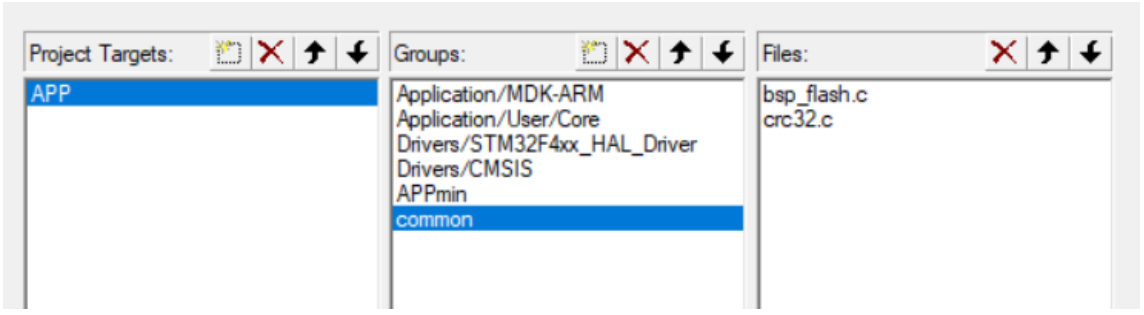



4. 创建工程

## 2.keil中文件的添加以及文件路径设置与代码添加

1. 添加文件如图所示




2. 点击魔法棒，添加头文件的路径(参考)-点击上面的Include Paths->然后添加刚刚创建好的BL_APP和Common文件的路径即可(**注意:一定要点进文件夹**)-操作与上方BL创建相同
3. 点开main.c补充代码

```
/* USER CODE BEGIN Includes */
#include "ota_uart_app.h"       //添加头文件定义
#define APP_BANNER  "APP v1.0 running\r\n"   // OTA 验证：下发前改成 "APP v2.0
/* USER CODE END Includes */
```

**主函数书写**

```
int main(void)
{

  /* USER CODE BEGIN 1 */
SCB->VTOR = 0x08014000UL;       //重点：防止中断跑飞
  /* USER CODE END 1 */

  /* MCU Configuration--------------------------------------------------------
-*/

  /* Reset of all peripherals, Initializes the Flash interface and the
Systick. */
  HAL_Init();

  /* USER CODE BEGIN Init */

  /* USER CODE END Init */

  /* Configure the system clock */
  SystemClock_Config();

  /* USER CODE BEGIN SysInit */

  /* USER CODE END SysInit */

  /* Initialize all configured peripherals */
  MX_GPIO_Init();
  MX_DMA_Init();
  MX_USART1_UART_Init();
  MX_USART3_UART_Init();
  /* USER CODE BEGIN 2 */
    ota_uart_start();
  HAL_UART_Transmit(&huart1, (uint8_t *)APP_BANNER, sizeof(APP_BANNER) - 1,
100);
  uint32_t t_ota = HAL_GetTick(), t_led = t_ota, t_beat = t_ota;
  /* USER CODE END 2 */

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
        uint32_t now = HAL_GetTick();
    if (now - t_ota  >= 5)    { t_ota  = now; ota_poll_task(); }
    if (now - t_beat >= 1000) { t_beat = now;
    HAL_UART_Transmit(&huart1, (uint8_t *)APP_BANNER, sizeof(APP_BANNER) - 1,
100); }
  }
  /* USER CODE END 3 */
}
```

4. **首先将Linker中取消勾选(如上方BL的图所示)->修改MDK_ARM文件夹下的APP.sct代码(修改或者直接复制我代码中APP.sct)->在Scatter File中选择修改好的APP.sct路径(重点)**

   **APP.sct代码**

```
LR_IROM1 0x08014000 0x00068000 {
  ER_IROM1 0x08014000 0x00068000 {
   *.o (RESET, +First)
   *(InRoot$$Sections)
   .ANY (+RO)
   .ANY (+XO)
  }
  RW_IRAM1 0x20000000 0x00020000 { .ANY (+RW +ZI) }
  ER_IRAM_FUNC +0 {
   bsp_flash.o (+RO)
  }
}
```

5. 编译下载->打开两个串口验证是否程序成功下载并运行(如图所示)

   OTA串口打印

   ```
   BL: jumping to app...
   ```

   测试串口打印

   ```
   [22:21:31.910]收←◆APP v1.0 running
   [22:21:32.917]收←◆APP v1.0 running
   [22:21:33.913]收←◆APP v1.0 running
   ```

# 上位机下发固件包以及单片机工程的远程更新

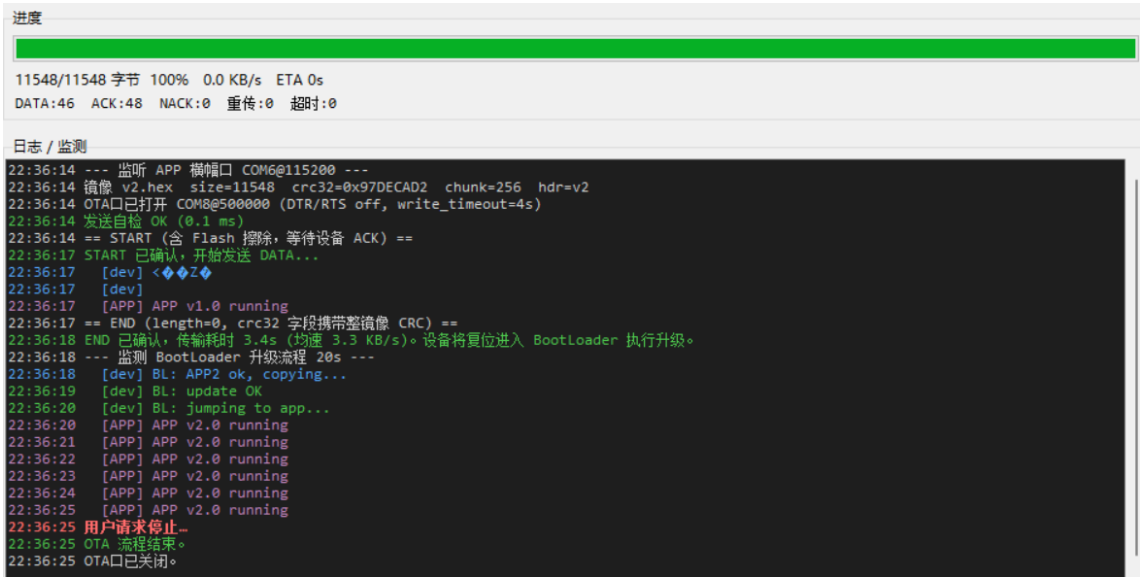
## 1.修改int main()主函数

1. 将APP v1.0 running->APP v2.0 running->重新编译**(不要进行下载)**
2. 将MDK_ARM文件夹里面的APP.hex复制粘贴出来，放置到随便的路径位置(你知道把文件放置到哪个位置就行)->将这个APP.hex改名为v2.hex(这一步可做可不做)

# 2.打开Tools文件夹运行ota_gui.py即可(注意下载需要的库)

1. 选择对应的OTA的串口，新固件的位置，测试串口，波特率大小(如图所示)


2. 点击开始升级->查看新固件是否成功下载并打印对应的输出(如图所示)




## 3.注意事项:烧录固件时推荐使用USB-TTL，否则部分固件有可能更新失败或者请求超时

## 恭喜您完成了bootloder的快速移植与使用，感谢您的学习